



SEA-BIRD
SCIENTIFIC

SBE37-SMP MicroCAT

Instrument Configuration

Instrument Serial Number: 37-16242
Instrument Firmware Version: 4.1
Zero Conductivity Frequency: 2833.26
Communications Format: RS232
Communications Settings: 9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

<i>Data Format</i>	<i>Measurement</i>	<i>Sensor Type</i>	<i>Serial Number</i>	<i>Rating</i>
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10723832	2000m(2000 dBar)

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.



Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 16242
CALIBRATION DATE: 03-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

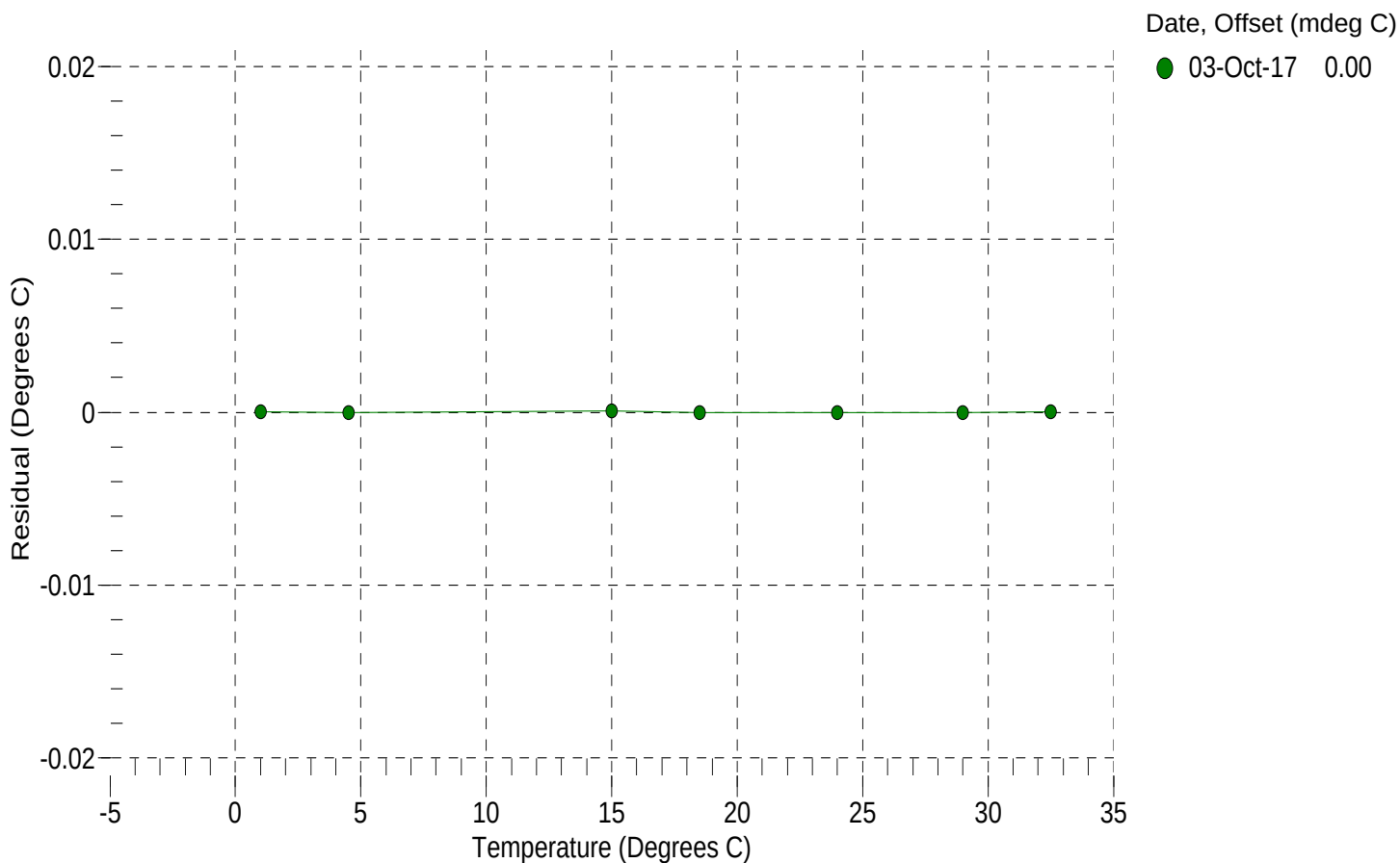
a0 = -2.404075e-004
a1 = 3.264361e-004
a2 = -5.525323e-006
a3 = 2.314773e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	557523.2	1.0000	0.0000
4.5000	478600.8	4.5000	-0.0000
15.0000	308642.7	15.0001	0.0001
18.5000	268293.0	18.5000	-0.0000
24.0000	216540.6	24.0000	-0.0000
29.0000	179281.0	29.0000	-0.0000
32.5000	157598.5	32.5000	0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature





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SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -1.014727e+000
h = 1.265057e-001
i = -9.807419e-005
j = 2.246157e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = 1.4705e-007

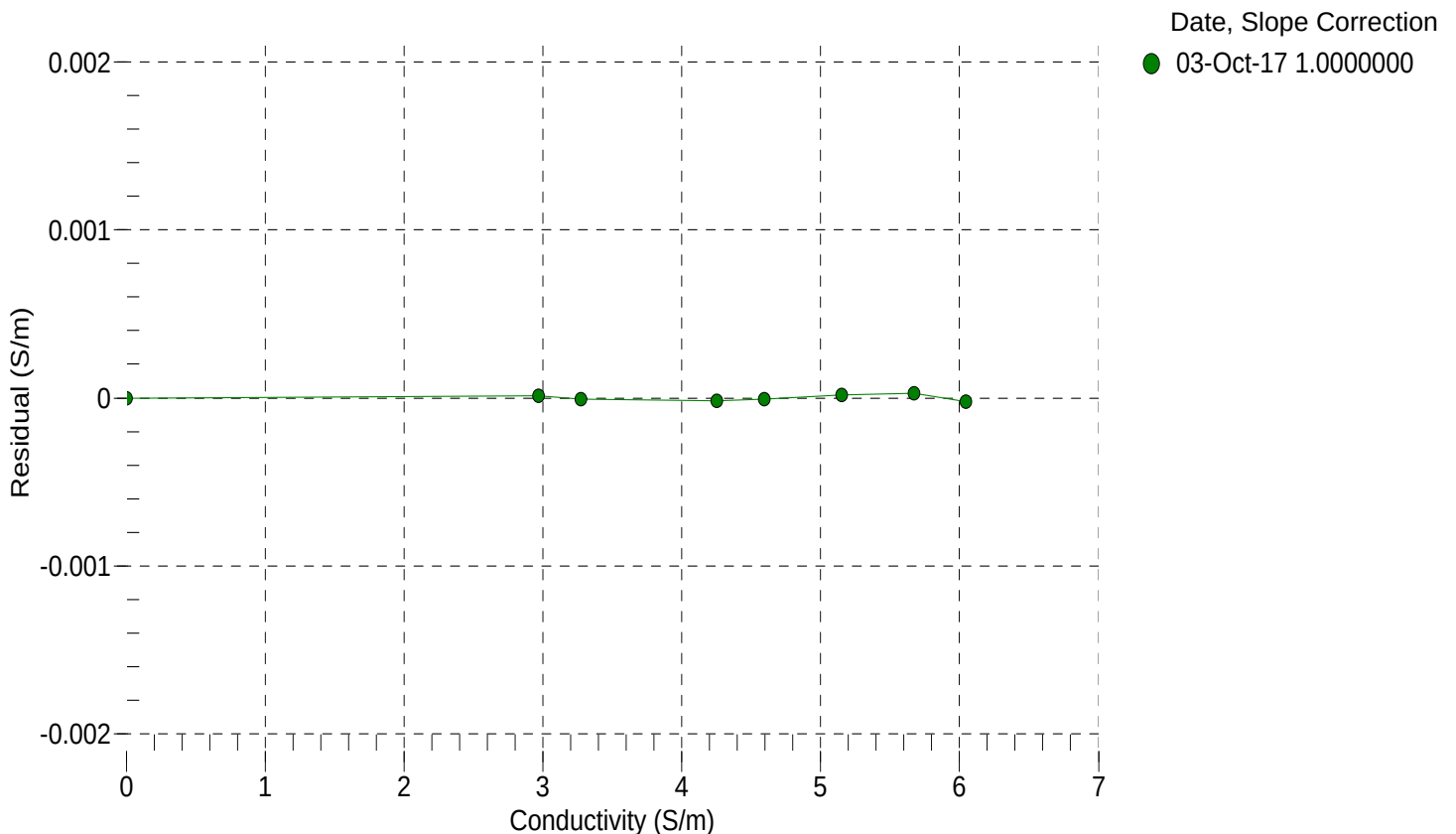
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2833.26	0.00000	0.00000
1.0000	34.6932	2.96644	5606.40	2.96646	0.00001
4.5000	34.6732	3.27255	5817.18	3.27254	-0.00001
15.0000	34.6306	4.25124	6444.30	4.25122	-0.00002
18.5000	34.6218	4.59534	6650.50	4.59534	-0.00001
24.0000	34.6123	5.15162	6970.67	5.15164	0.00002
29.0000	34.6073	5.67193	7257.04	5.67195	0.00003
32.5000	34.6050	6.04330	7454.50	6.04328	-0.00002

$f = \text{Instrument Output(Hz)} * \sqrt{1.0 + \text{WBOTC} * t} / 1000.0$

t = temperature (°C); p = pressure (decibars); $\delta = \text{CTcor}$; $\epsilon = \text{CPcor}$;

$\text{Conductivity (S/m)} = (g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \epsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity





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SENSOR SERIAL NUMBER: 16242
CALIBRATION DATE: 27-Sep-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10723832

COEFFICIENTS:

PA0 =	4.900158e-001	PTCA0 =	5.242409e+005
PA1 =	9.112524e-003	PTCA1 =	3.591802e+000
PA2 =	-1.477475e-010	PTCA2 =	-5.547330e-002
PTEMPA0 =	-5.673827e+001	PTCB0 =	3.064031e+005
PTEMPA1 =	5.379882e-002	PTCB1 =	-2.752003e+001
PTEMPA2 =	-6.564868e-007	PTCB2 =	1.316803e+000

PRESSURE SPAN CALIBRATION

THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
14.63	525855.1	1512.8	14.71	0.00	32.50	1694	525933.35
591.69	589241.5	1514.4	591.56	-0.00	29.00	1626	525936.19
1168.83	652801.9	1514.6	1168.80	-0.00	24.00	1529	525934.22
1746.03	716494.0	1514.9	1746.05	0.00	18.50	1423	525924.75
2323.18	780307.7	1515.1	2323.19	0.00	15.00	1356	525916.05
2900.19	844228.9	1515.6	2900.09	-0.00	4.50	1155	525891.94
2323.21	780324.3	1514.8	2323.34	0.00	1.00	1088	525881.67
1746.04	716498.4	1514.0	1746.09	0.00	TEMPERATURE (°C) SPAN (mV)		
1168.98	652822.0	1513.0	1169.00	0.00			
591.65	589239.4	1512.6	591.55	-0.00			
14.62	525851.9	1512.4	14.68	0.00			
					4.32	306308.88	
					22.95	306465.05	
					34.50	307021.22	

y = thermistor output (counts)

t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y²

x = instrument output - PTCA0 - PTCA1 * t - PTCA2 * t²

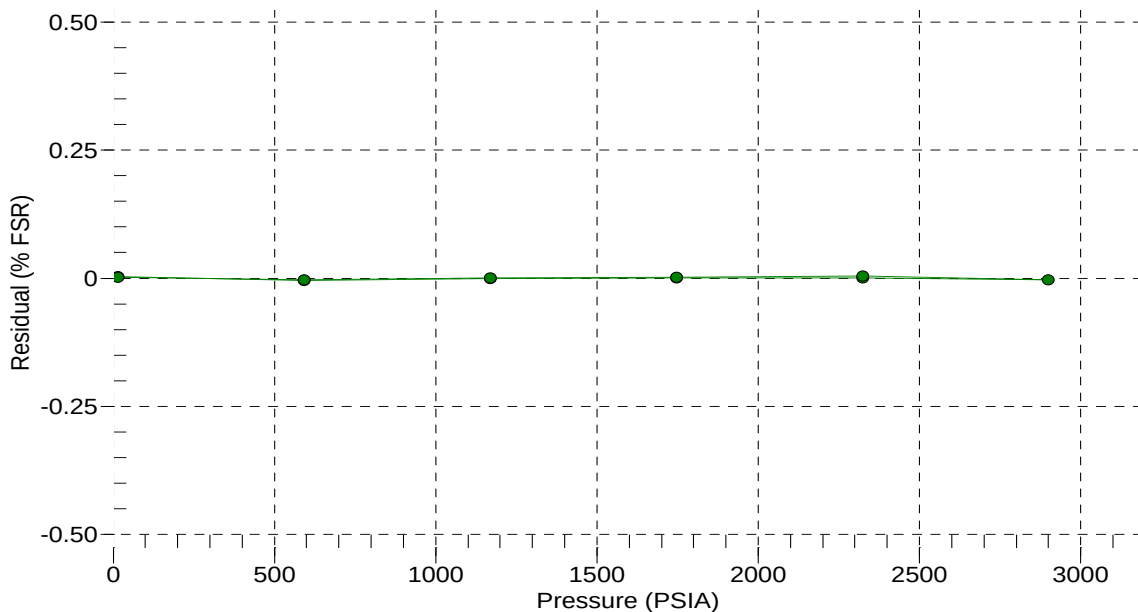
n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t²)

pressure (PSIA) = PA0 + PA1 * n + PA2 * n²

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range

Date, Offset (%FSR)

● 27-Sep-17 -0.00





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Pressure Test Certificate

Test Date: **2017-10-09**

Description: **SBE-37 Microcat**

Sensor Information:

Model Number: **SBE-37**

Serial Number: **16242**

Pressure Test Protocol:

Low Pressure Test: **40** PSI Held For: **15** Minutes

High Pressure Test: **2900** PSI Held For: **15** Minutes

Passed Test: **True**

Tested By: **TH**

